Design Proposal for Mach–Zehnder Interferometer Course Project

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***Abstract*—This paper presents the theoretical analysis, design, and numerical simulation of Mach–Zehnder interferometers (MZIs) implemented on a silicon-on-insulator (SOI) platform. Silicon strip waveguides are designed for both single-mode and multimode operation using numerical eigenmode analysis, and their modal characteristics, dispersion properties, and bending losses are systematically investigated. Compact models of the effective refractive index are developed. The simulated transmission spectra show free spectral ranges that are in good agreement with analytical predictions, validating the proposed modeling and design methodology.**

I. Introduction

Silicon photonics has emerged as a key enabling technology for next-generation optical communication systems, driven by its compatibility with complementary metal–oxide–semiconductor (CMOS) fabrication processes, high integration density, and low power consumption [1], [2]. In particular, photonic integrated circuits (PICs) based on silicon-on-insulator (SOI) platforms have been widely adopted for high-speed optical interconnects in data-center and telecommunication applications [3], [4].

Among various photonic building blocks, the Mach–Zehnder interferometer (MZI) plays a central role in silicon photonics. MZIs serve as fundamental components for a wide range of functionalities, including optical modulation, switching, filtering, and phase-sensitive signal processing [5], [6]. Their broadband operation, high extinction ratio, and design flexibility make MZIs indispensable in practical transceiver architectures for optical communication systems.

Beyond conventional communication applications, MZIs have recently attracted significant attention in emerging computing paradigms such as optical accelerators and neuromorphic computing. In photonic neural networks and optical matrix–vector multiplication engines, MZIs are widely used as programmable linear transformation units that implement weighted interference and phase control [7], [8], [9]. The scalability and reconfigurability of MZI meshes have positioned them as core architectural elements for optical computing and artificial intelligence hardware accelerators.

As the scale and functionality of MZI-based circuits continue to increase, challenges related to device footprint, loss, phase errors, and fabrication tolerance become increasingly critical. In particular, compact and low-loss MZI designs are essential to enable large-scale integration while maintaining energy efficiency and signal fidelity. These requirements motivate the exploration of new MZI design methodologies optimized for both optical performance and physical compactness.

II. Theory of the Mach–Zehnder Interferometer

The ideal behavior of a MZI can be analytically described by deriving its transfer function. In this section, we formulate the transfer function of an ideal MZI under commonly adopted assumptions. The splitter and combiner stages of an MZI are typically implemented using either a 1×2 multimode interference (MMI) coupler, a Y-branch, or a directional coupler (DC). Figure 1 illustrates a representative MZI configuration employing Y-branches as both the splitter and the combiner. In general, phase modulation in an MZI is introduced by creating an imbalance between the two arms. This can be achieved either by introducing a path length difference ΔL between the two arms or by modulating the propagation constant β via an effective refractive index change. Such configurations are commonly referred to as imbalanced MZIs.

We assume that the input Y-branch ideally splits the optical power equally into the two arms. If the incident optical intensity and field amplitude are denoted by I_i and E_i, respectively, the optical intensities in the two arms are given by

$$I_1 = I_2 = \frac{I_i}{2}. \tag{1}$$

Correspondingly, the electric field amplitudes in each arm are

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}}. \tag{2}$$

Let the lengths of the two arms be L_1 and L_2, respectively. The corresponding propagation constants are denoted by

$$\beta_k = \frac{2\pi n_{\mathrm{eff},k}}{\lambda} \quad (k = 1, 2), \tag{3}$$

where $n_{\mathrm{eff},k}$ is the effective refractive index of the k-th arm and λ is the vacuum wavelength. We also introduce the power attenuation coefficients α_1 and α_2 for the two arms. The complex phase accumulation during propagation can be written as

$$\exp\left[-i\left(\beta_k - i\frac{\alpha_k}{2}\right)L_k\right]. \tag{4}$$

Therefore, the electric field at the end of each arm is expressed as

$$E_{o,k} = \frac{E_i}{\sqrt{2}} \exp\left[-i\left(\beta_k - i\frac{\alpha_k}{2}\right)L_k\right], \quad k = 1, 2. \tag{5}$$

When the two optical fields recombine at the output Y-branch

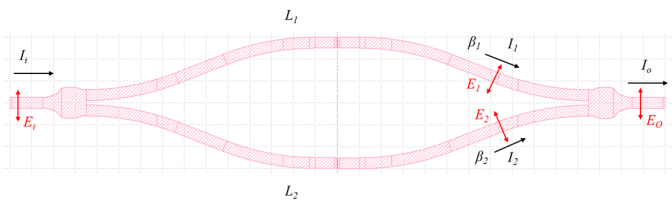

Fig. 1. Schematic of a Mach–Zehnder interferometer employing Y-branches as the splitter and combiner. An optical path length difference ΔL between the two arms, L_1 and L_2 induces wavelength-dependent interference at the output.

(or combiner), each field is again split between the fundamental mode and higher-order or radiation modes. Assuming ideal recombination into the fundamental mode, an additional factor of $1/\sqrt{2}$ applies to each arm. The total output electric field is thus given by the sum of $E_{o,1}$ and $E_{o,2}$ as

$$E_o = \frac{E_i}{2}\Big[\exp\left(-i\left(\beta_1 - i\frac{\alpha_1}{2}\right)L_1\right) + \exp\left(-i\left(\beta_2 - i\frac{\alpha_2}{2}\right)L_2\right)\Big]. \tag{6}$$

In the special case where both arms have identical propagation constants and absorption coefficients, i.e.,

$$\beta_1 = \beta_2 = \beta, \quad \alpha_1 = \alpha_2 = \alpha, \tag{7}$$

the output intensity becomes

$$I_o = \frac{I_i}{4}\left|\exp\left(-i\left(\beta - i\frac{\alpha}{2}\right)L_1\right) + \exp\left(-i\left(\beta - i\frac{\alpha}{2}\right)L_2\right)\right|^2. \tag{8}$$

After simplification, this yields

$$I_o = \frac{I_i}{2}\left[\frac{e^{-\alpha L_1} + e^{-\alpha L_2}}{2} + e^{-\alpha(L_1+L_2)/2}\cos(\beta\Delta L)\right], \tag{9}$$

where the path length difference is defined as

$$\Delta L = |L_1 - L_2|. \tag{10}$$

In the limit of negligible absorption ($\alpha \approx 0$), the output intensity simplifies to

$$I_o = \frac{I_i}{2}\left[1 + \cos(\beta\Delta L)\right]. \tag{11}$$

The resulting periodic modulation corresponds to the free spectral range (FSR). Expressed in terms of the propagation constant β, the FSR is given by

$$\mathrm{FSR}(\beta) = \frac{2\pi}{\Delta L}. \tag{12}$$

Using the relation $\beta = 2\pi n_{\mathrm{eff}}/\lambda$ and introducing the group index

$$n_g = n_{\mathrm{eff}} - \lambda\frac{dn_{\mathrm{eff}}}{d\lambda}, \tag{13}$$

the FSR in the wavelength domain is obtained as

$$\mathrm{FSR}(\lambda) = \frac{\lambda^2}{n_g\Delta L}. \tag{14}$$

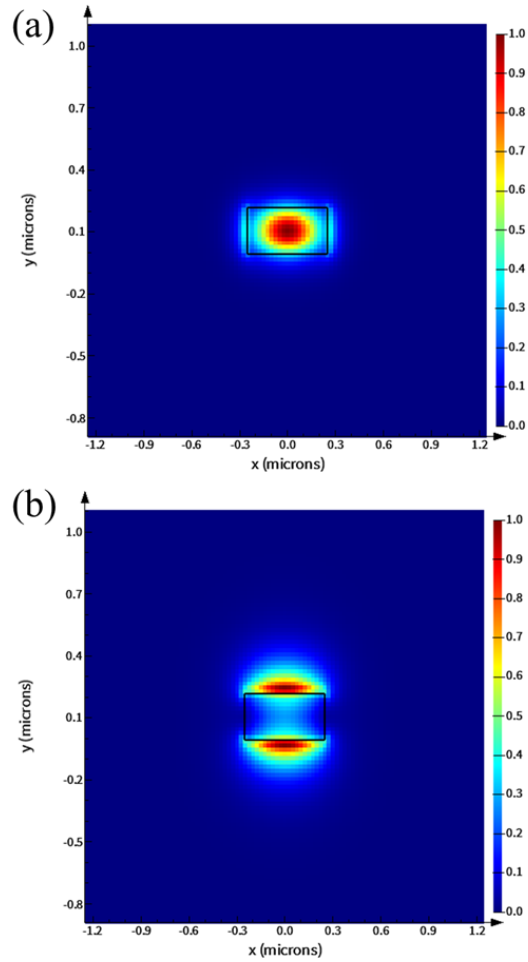

Fig. 2. Simulated optical field distributions of the fundamental (a) TE-like and (b) TM-like modes for the 500 nm-wide waveguide.

III. Modeling and Simulation

A. Waveguide Design

In this section, we describe the design overview of the Si waveguides used in this study. Strip-type silicon waveguides were employed, and their dimensions were chosen to ensure single-mode operation. Specifically, the waveguide width and thickness were set to 500 nm and 220 nm, respectively. The waveguide eigenmodes and their optical properties were calculated using the MODE eigenmode solver in the Lumerical simulation suite. Figure 2 shows the simulated optical field distributions of the fundamental transverse-electric-like (TE-like) and transverse-magnetic-like (TM-like) modes for the designed waveguide.

As the waveguide width increases, higher-order modes such as the TE_2, TM_2 are supported in addition to the fundamental modes. When the waveguide width approaches approximately 900 nm, a distinct TE_2 mode with two well-separated lobes is clearly observed. Even at a width of 600 nm, although the mode profile does not exhibit a typical TE_2-like pattern, modes with an effective refractive index exceeding that of the SiO_2

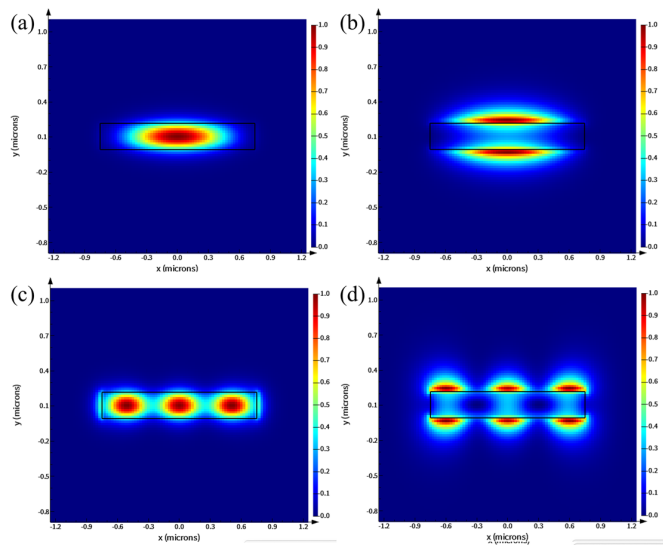

Fig. 3. Simulated optical mode profiles of the fundamental (a) TE_1 and (b) TM_1 modes, and higher-order (c) TE_3 and (d) TM_3 modes for the 1.5 µm-wide waveguide.

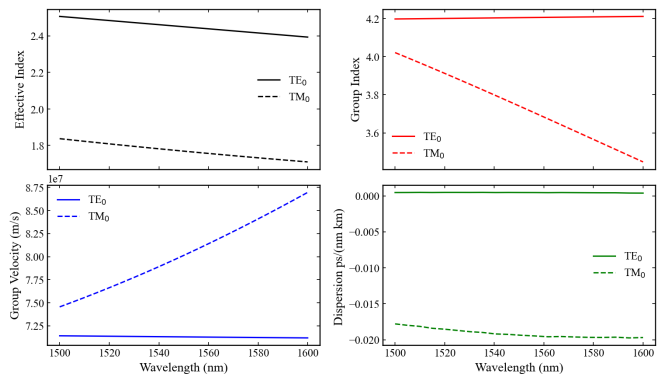

Fig. 4. Wavelength dependence of key optical properties for the single-mode waveguide with a width of 500 nm.

cladding ($n \approx 1.44$) are present.

This indicates a potential transition to multimode operation. Therefore, a waveguide width of 500 nm was selected as the baseline design to ensure robust single-mode operation. To mitigate scattering losses arising from sidewall roughness, an additional multimode waveguide design with an increased width of 1.5 µm was also considered. For this width, higher-order modes such as TE_3 and TM_3, characterized by three-lobed field distributions, are supported. The simulated mode profiles for the 1.5 µm-wide waveguide are shown in Figure 3.

Figure 4 and Figure 5 summarize the wavelength dependence of the effective refractive index, group index, group velocity, and dispersion for the 500 nm-wide and 1.5 µm-wide waveguides, respectively. The chromatic dispersion is expressed using the group-velocity dispersion (GVD) parameter β_2 as

$$D = -\frac{2\pi c}{\lambda^2}\beta_2, \tag{15}$$

where c is the speed of light in vacuum. The GVD is related

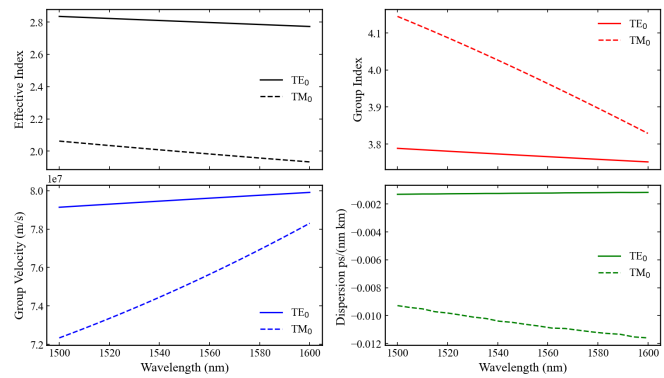

Fig. 5. Wavelength dependence of key optical properties for the multimode waveguide with a width of 1.5 µm.

TABLE I
FITTING PARAMETERS OF THE COMPACT MODEL FOR THE 500 NM WIDE SINGLE-MODE WAVEGUIDE.

Parameter	TE_0	TM_0
n_0	2.4502	1.7692
n_1	-1.1320	-1.2721
n_2	-0.04416	1.8616

to the group velocity v_g as

$$\beta_2 = \frac{d}{d\omega}\left(\frac{1}{v_g}\right), \tag{16}$$

with the group velocity given by

$$v_g = \frac{c}{n_g}, \tag{17}$$

where n_g denotes the group refractive index. Consequently, the dispersion is directly proportional to the wavelength derivative of the group refractive index.

A compact models for the effective refractive index for the single-mode and multimode waveguides were constructed by performing a least-squares fit to the simulated effective index data shown in Fig. 4 and Fig. 5, respectively. The effective refractive index was approximated by a second-order Taylor expansion around a reference wavelength $\lambda_0 = 1550$ nm as

$$n_{\text{eff}}(\lambda) = n_0 + n_1(\lambda - \lambda_0) + n_2(\lambda - \lambda_0)^2. \tag{18}$$

The coefficient of determination (R^2) was found to be approximately unity, indicating excellent agreement between the compact model and the numerical simulation. The extracted fitting parameters for the single-mode and multimode waveguides are summarized in Table I and Table II, respectively.

TABLE II
FITTING PARAMETERS OF THE COMPACT MODEL FOR THE 1.5 µM-WIDE MULTIMODE WAVEGUIDE.

Parameter	TE_0	TM_0
n_0	2.8029	1.9953
n_1	-0.6237	-1.2895
n_2	0.1183	1.0223

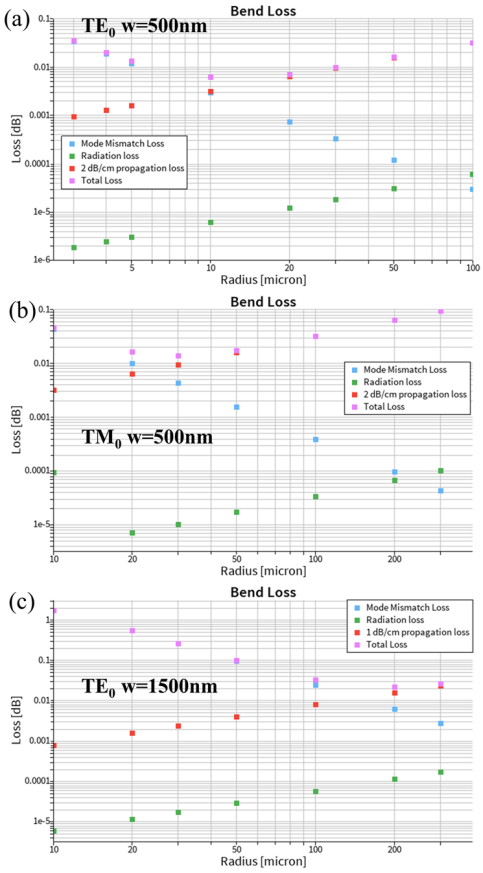

Fig. 6. Simulated bending loss as a function of bending radius. (a) TE_1 mode and (b) TM_1 mode in the single-mode waveguide. (c) Multimode waveguide with a width of 1.5 μm. The bending loss is evaluated based on the squared overlap integral between the field distributions of straight and bent waveguides, including the assumed contributions of material absorption and scattering losses.

B. Bending Radius Design

The bending radius was investigated for both single-mode and multimode waveguides. The bending loss was evaluated by calculating the squared overlap integral of the electric field distributions between a straight waveguide and a bent waveguide, which represents the loss induced by mode mismatch. Although radiation loss due to outward leakage in the radial direction is also present, the dominant contribution to the bending loss originates from mode mismatch in the present waveguide configurations.

In addition to the bending-induced loss, material absorption and scattering losses were assumed to be constant and independent of the bending radius. These contributions were set to 2 dB/cm for the single-mode waveguide and 1 dB/cm for the multimode waveguide. The simulation results are summarized in Fig. 6.

To suppress the total bending loss to a negligible level on the order of 0.01 dB or lower, the bending radii were determined based on the results shown in Fig. 6(a) and Fig. 6(b). For the single-mode waveguide, bending radii of $r = 7$ μm and $r = 15$ μm were adopted for the TE_1 and TM_1 modes, respectively.

TABLE III
CALCULATED GROUP INDEX n_g AT $\lambda_0 = 1550$ NM DERIVED FROM THE COMPACT MODEL FITTING COEFFICIENTS.

Waveguide	TE_0	TM_0
500 nm (single-mode)	4.20	3.74
1.5 μm (multimode)	3.77	3.99

TABLE IV
CALCULATED FSR AT $\lambda_0 = 1550$ NM FOR DIFFERENT PATH LENGTH DIFFERENCES ΔL.

	FSR (nm)			
ΔL (μm)	TE_0 (500 nm)	TM_0 (500 nm)	TE_0 (1.5 μm)	TM_0 (1.5 μm)
25	22.9	25.7	25.5	24.1
50	11.4	12.9	12.7	12.1
100	5.71	6.43	6.36	6.03
200	2.86	3.21	3.18	3.01
400	1.42	1.61	1.59	1.51

For the multimode waveguide with a width of 1.5 μm, as shown in Fig. 6(c), the increased waveguide width leads to a larger mode mismatch at bends, resulting in higher bending loss. In this case, a bending radius of $r = 80$ μm was selected to sufficiently suppress the bending-related loss.

C. MZI Design

From the compact model, the group index n_g at the reference wavelength $\lambda_0 = 1550$ nm can be analytically derived from the fitting coefficients, given by

$$n_g(\lambda_0) = n_0 - \lambda_0 n_1. \tag{19}$$

. The group indices for the single-mode and multimode waveguides were calculated from the fitted coefficients listed in Tables I and II. The resulting group indices at $\lambda_0 = 1550$ nm are summarized in Table III.

The FSR of the MZI was calculated using Eq. (14). Table IV summarizes the calculated FSR values for different path length differences ΔL for both single-mode and multimode waveguides.

MZI simulations were carried out using the EBeam process design kit (PDK) provided by the University of British Columbia (UBC). The simulations were performed with the INTERCONNECT module of the Lumerical simulation suite. Y-branches were employed as both the splitter and the combiner, and simulations were conducted for single-mode waveg-

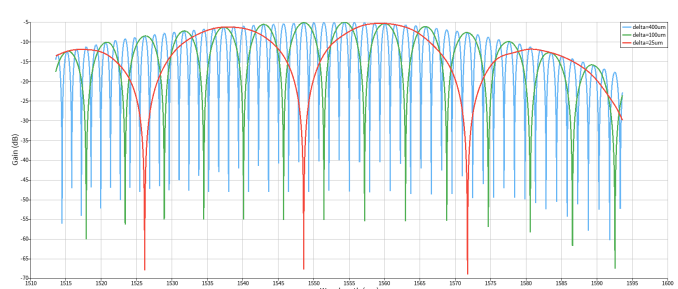

Fig. 7. Simulated transmission spectra of MZIs designed using the EBeam PDK. The path length differences were $\Delta L = 25$, 100, and 400 μm.

uides with path length differences of $\Delta L = 25$, 100, and 400 μm. All simulations focused on the TE polarized mode.

The simulated transmission spectra for the TE mode are shown in Fig. 7. For example, the FSRs around the central wavelength $\lambda_0 = 1550$ nm were found to be 1.4 nm, 5.6 nm, and 21.6 nm for $\Delta L = 25$, 100, and 400 μm, respectively. These values are in good agreement with the analytical predictions based on the expressions summarized in Table IV.

IV. Conclusion

Under construction.

References

[1] R. Soref, "The past, present, and future of silicon photonics," *IEEE J. Sel. Top. Quantum Electron.*, vol. 12, no. 6, pp. 1678–1687, 2006.

[2] D. Thomson, A. Zilkie, J. E. Bowers, *et al.*, "Roadmap on silicon photonics," *J. Opt.*, vol. 18, 073003, 2016.

[3] C. Sun *et al.*, "Single-chip microprocessor that communicates directly using light," *Nature*, vol. 528, pp. 534–538, 2015.

[4] D. A. B. Miller, "Attojoule optoelectronics for low-energy information processing and communications," *J. Lightwave Technol.*, vol. 35, no. 3, pp. 346–396, 2017.

[5] M. J. R. Heck and J. E. Bowers, "Energy efficient and energy proportional optical interconnects for multi-core processors," *IEEE J. Sel. Top. Quantum Electron.*, vol. 20, no. 4, pp. 1–12, 2013.

[6] W. Bogaerts *et al.*, "Programmable photonic circuits," *Nature*, vol. 586, pp. 207–216, 2020.

[7] Y. Shen *et al.*, "Deep learning with coherent nanophotonic circuits," *Nature Photonics*, vol. 11, pp. 441–446, 2017.

[8] J. Feldmann *et al.*, "All-optical spiking neurosynaptic networks with self-learning capabilities," *Nature*, vol. 569, pp. 208–214, 2019.

[9] A. N. Tait *et al.*, "Neuromorphic photonic networks using silicon photonic weight banks," *Sci. Rep.*, vol. 9, 8033, 2019.